

PH3103 Mathematical Methods of Physics
Autumn Semester - 2025
Indian Institute of Science Education and Research, Kolkata
Instructor: Koushik Dutta

Homework: 11

Submission Date: 20/11/2025

The hand written solutions must be submitted at the start of the class

1. Find out the Green's function for the diffusion operator in one space dimension x :

$$\left(\frac{\partial}{\partial t} - a^2 \frac{\partial^2}{\partial x^2} \right) G(x, t; x', t') = \delta(x - x') \delta(t - t') . \quad (1)$$

Assume boundary conditions are translationally invariant.

2. Find the iterative solution for the following equation up to $\mathcal{O}(\lambda^3)$:

$$y(t) = x + \frac{\lambda}{2} \int_{-1}^1 (t - x) y(x) dt \quad (2)$$